

I Learned Architecture Before Machines Learned to Imagine It

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ABSTRACT

An architect who learned the discipline before machines learned to imagine it tells the story of the collapse of his profession's gatekeeping, and argues it is good news. Opening with the twenty-five-year biography of a single material (exposed concrete: loved for its truth, rejected as unfinished, and now sold as a peel-and-stick picture of itself), the essay traces the industrialization of an old pattern the Bilbao Effect literature knows well: the market believes the render is the promise. Drawing on his own years as a prolific renderer, a hospital-bed first encounter with image generation that arrived as rescue rather than threat, and current adoption data showing acceleration without improvement, the author proposes four laws for what replaces the priesthood: instruments of judgment built on constraint, honest math, refusal, and accountability as the product, declaring his own stake in building them, and arguing the laws hold for his competitors too.

"They took the importance away from thinking and gave it to concrete." The author, Sobreviviendo a la Ciudad, 2015

In the early 2000s I fell in love with exposed concrete. I had seen it in magazines, in books, in the courses where they teach you what architecture is supposed to want, and when I finally cast my first wall in the real world I understood the photographs had been underselling it: the texture, the way light drags across it in the afternoon, the weight. I tried it a few times. My clients looked at the same wall and saw something else: an unfinished surface, waiting for plaster. I stopped offering it.

In 2016 I tried again: metal molds, cast-in-place, a row of townhouses I believed in. I could not sell them.

Then came the pandemic, and after it, a small miracle of the global supply chain: today you can order PVC panels from Temu, peel off the backing, and stick the *image* of exposed concrete to any wall in your house. They look like my wall from 2003. They feel like a shower curtain. The market that once rejected the material as unfinished now buys its picture by the twelve-pack.

Last month I spent some weeks in Japan, and put my hand on Tadao Ando's concrete, and believed the material again.

I am telling you the biography of one wall because it is also the biography of my profession. Something we loved for its truth was rejected for its honesty, and then returned as a sticker. That is, in one wall, everything that has happened to architecture since machines learned to imagine.

The machines learned fast. This July, the Royal Institute of British Architects reported that 74 percent of UK practices now use AI on most of their projects, fifteen points more than a year ago; the image generators produce tens of millions of pictures a day; in visualization studios, the number-one use of the new tools is generating concept images. And inside the same survey sits the number nobody reads aloud: only 17 percent of those practices say their designs got better. Faster, almost all of them. Better, one in six.

None of this should surprise anyone who watched the icon decades. The discourse has a name for what happens when a city buys an image instead of a judgment, the Bilbao Effect, and twenty years of scholarship on it reads like a coroner's report: one miracle in Bilbao, then a generation of expensive imitations catalogued as narrative first, results later. I wrote my own small chapter of that literature in 2015, about my hometown's municipal white elephants. The epigraph above is what I concluded, watching public money pour into objects that, as I wrote then, only shine on inauguration day.

The pattern was already old when the machines industrialized it. **The market believes the render is the promise.** It believed it in Bilbao's imitators, it believed it in my town's plazas, and it believes it now at a scale of millions of images a day.

I am allowed to say this because I was one of the believers, and I was armed. Between 2005 and 2009 I committed renders at a pace that would embarrass nobody in the Midjourney era. I had learned 3D in high school; I hacked Bryce, a piece of gaming software, to get ahead of the archaic 3ds Max versions of the day; SketchUp early, Cinema 4D, later Blender and Twinmotion. Rendering was never the hard part for me. And I believed, naively, that more renders would mean more contracts, and yes, they did. What I had not understood is that contracts are made of responsibilities, and inside the responsibilities the renders have no room. A contract is a kitchen

that must pass inspection, a slab that must carry water it will never be photographed carrying, a neighbor with a lawyer, a change order signed at seven in the evening. Ask yourself what any image, however beautiful, contributes at that table.

All those years of rendering bought me exactly one thing, and it turned out to be the valuable one: the eye. I can look at a beautiful image and know it is not a beautiful building, the way a butcher looks at a photograph of a cow. What the buildings themselves taught me was different and slower: that a building has veins, systems under the skin, and that the education which begins at the drafting table only finishes inside a mechanical room. That second education is the one I now spend on food-and-beverage facilities on an industrial scale, where nobody has ever once asked me for a mood board.

And I want to be exact about the machines, because my first real night with them was not a threat. It was a rescue. In the worst season of my consulting years I was flying three times a week across three time zones, carrying a prototype for automated indoor farming that had sat unshown for months because there was never a week to make the visuals. Then an emergency landing, and a hospital bed, and my own human system malfunctioning. And that night, in the bed, I found the image generator, and by morning I had everything I had been unable to make: months of imagination, poorly represented but *there*. I have never been anti-machine since. I am something more demanding than that.

Because here is what the hospital also taught me: the imagination was recovered in a night. The judgment of what to do with it had taken twenty years, and it was lying in the bed too. Experience that stays in a person retires with the person. The question of my generation is how judgment becomes infrastructure.

I do not think the answer is defending the priesthood. My profession spent a century wrapping its method in credentials and mystique, and the people it priced out deserved better than gatekeeping: the small towns, the first-time owners, the modest budgets. The gates are down. Good. The real question is what a serious instrument of judgment looks like when anyone can generate the image. I should declare my stake plainly: I am building such an instrument, so discount everything in this section accordingly. But I will argue the laws on their merits, because I believe they hold for anyone who builds one, including my competitors.

First, constraint instead of an open prompt. A serious instrument does not ask the client to invent; it offers what is known (real building types, curated options, structure that unfolds), because the open prompt hands the burden of judgment back to the person who came to buy it. Second, honest math: your budget carries eleven rooms, not the sixteen in the picture, and here is the range with its sources: the truth about the project before the wish about the project. Third, refusal. What the machine declines to estimate, and the confidence it declines to claim when the human gave it nothing, is where its integrity lives; an empty field is honest, a wrong number is malpractice. And fourth, accountability as the product: let the knowledge flow free to

anyone who scrolls, and reserve the price for what someone must stand behind: the signed number, the document that can be executed against. The teenager exploring at midnight learns; the investor committing millions gets a signature. Both are treated with dignity.

Nothing in those four laws requires my generation. But writing them requires having been at the table where the render died, and that table is the one thing the machines have never attended.

The obvious objection is generational vanity, and half of it is right. The young will acquire judgment too, and faster than we did, with better tools. Every generation has called the next one shallow, and every generation has been wrong. I concede it happily. The half that fails is arithmetic. Judgment is learned from consequences, and consequences did not accelerate: the models retrain in weeks, a permit still takes a year, a hotel three, a lawsuit five. The bottleneck moved into the world, and the world does not do batch processing. Nor does hiring the gray-haired solve it: a person's judgment scales to one calendar and retires with one body. Persons retire. Systems compound. And I know the limits of my own inheritance from paying for it: I watched a hotel I worked on close because its region's security collapsed while the country's tourism grew, and the townhouses I still believe in never sold. Scar tissue is not foresight. Experience prices risk, which is exactly why it belongs inside systems that state their confidence honestly, instead of inside auras that cannot.

Be clear about who wins if we do this honestly. Somewhere in Queens tonight, a teenager is producing a competent brief for a small hotel on a street she walks every day, with tools that arrived before the gatekeepers did. For her, the collapse of the priesthood is the first fair thing that has ever happened in this profession. The promise is real: knowledge, finally free. The trap is one substitution away: free knowledge mistaken for free judgment, a hundred beautiful options and no way to know which one survives a budget. An instrument that tells her the truth about her street costs her nothing and teaches her everything.

For the profession, the repricing is already visible. When the image was scarce, the image was the product; now that it is free, it is worthless as evidence, and what the market will pay for shifts to what cannot be generated: execution, systems, a signature that survives contact with a general contractor. There is a window for the people who understand consequences to define what the AI-native version of this profession becomes, and windows like it close early. Categories get named after whoever ships first, and the people shipping fastest right now have never carried a building. The market believes the render is the promise. It will keep believing that until professionals show it something better to believe in.

I learned architecture before machines learned to imagine it, and I am glad I did, and I am glad they did. The wall I loved in 2003 is still standing; the plaster my clients wanted has been repainted four times. The render was never the promise. The building was: the veins behind the

wall, the kitchen that passes inspection, the number someone signs. The work now is to pour what we learned, like the material I never stopped loving, into instruments that carry its texture, its light, and its weight.

We are casting the priesthood's replacement. This time the market will get to touch it.

Jair Ibarra is a Mexican architect and founder of Rooterhub, an AI architecture and hospitality intelligence practice. He is the author of Sobreviviendo a la Ciudad (2015).

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